Identities and expressions on one page



Hi! Here is the revision before the test: identities, using a "block", the smallest value, comparing powers and proofs.

Whole expressions 1

A **whole expression** is built from numbers and letters with +, −, ×, powers and division **by a number**.

whole expressions	not whole
$3a^2 - 5b$	$5 \div x$
$(x + 1) \div 2$	$7 \div (a + b)$
$0.5m^3 + 7$	$a^2 \div b$

divide by a letter: not whole any more

A monomial is the simplest whole expression: multiplication only.

An identity 2

An **identity** is an equation that is true for **every** value of the letters. For example, $2(a + 3) = 2a + 6$.

a	$2(a + 3)$		$2a + 6$
1	8	=	8
2	10	=	10
5	16	=	16

the same for any a!

In UK textbooks an identity is written with the sign ≡.



A table is only a check. You prove it by changing the expression.

Key rule: how to prove an identity 3

Change one side step by step until it looks exactly like the other. ① Expand the brackets. ② Collect like terms.



Step by step, no jumps, and nobody can doubt it!

$$3(x - y) + 2(x + y) - x$$

↓ ① *expand the brackets*

$$3x - 3y + 2x + 2y - x$$

↓ ② *collect like terms*

$$4x - y$$

$3x + 2x - x = 4x$

$-3y + 2y = -y$

One counterexample is enough 4

To show an equation is **not** an identity, one set of numbers where it fails is enough.

$$(a + b)^2 = a^2 + b^2 \ ?$$

try $a = 1$, $b = 1$

$$(1 + 1)^2 = 4 \neq 1^2 + 1^2 = 2$$

$4 \neq 2$: not an identity!



I thought $(a + b)^2 = a^2 + b^2$. One example, and it's clear!

Use the block 5

Know the value of a whole "block"? Build the expression you need from it: multiply it or raise it to a power.

$2ab = 7$

$6ab = 3 \times (2ab) = 3 \times 7 = 21$

$4a^2b^2 = (2ab)^2 = 7^2 = 49$

no need to find a and b!

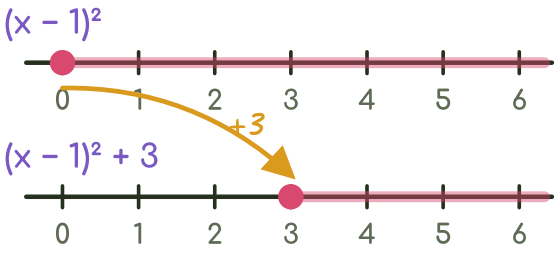
To spot the block, split the numbers: $4 = 2^2$, $6 = 3 \times 2$.

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The smallest value

A square (and any even power) is never negative: $x^2 \geq 0$. So $(x - 1)^2 + 3 \geq 3$.

The smallest value happens when the bracket is zero: $x = 1$.



smallest value 3, when x = 1

Comparing powers

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Rewrite the powers with the **same index** (or the same base), and comparing becomes easy.

$$2^{15} \;?\; 3^{10}$$

$$(2^3)^5 = 8^5 \qquad (3^2)^5 = 9^5$$

$$8^5 < 9^5$$

same index? compare the bases

So $2^{15} < 3^{10}$. Check: 32,768 < 59,049.

Proving divisibility

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The sum of three consecutive whole numbers is divisible by 3. We prove it for **any** n, not just a few examples.

$$n + (n + 1) + (n + 2)$$

$$= 3n + 3$$

$$= 3(n + 1)$$

a factor of 3!

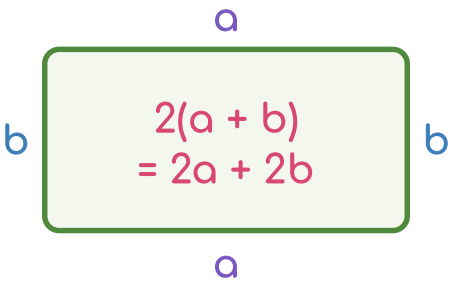
We took out the common factor 3, so it divides by 3.

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Identities in real life

A fence round a pen: you can go "length + width" twice, or add two lengths and two widths. Same answer: that's an identity.

Identities let you choose the easier way and be sure the answer won't change.





Check yourself: put one or two numbers into your answer. Don't match? Look for the mistake. Match? Prove it by rewriting.

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Built on his slides and lessons. Made by Mykola Stepanov for his daughter Varia